

EXERCISES AND EXACT SOLUTIONS

See the Service Gaps: Informal-Settlement Mapping

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C53
Notebook	V7_D_N03
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Replace straight-line distance with a network measure.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Add uncertainty ranges to coverage estimates.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Design a community-validation protocol.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 4

Explain why “informal” is not an analytical risk label.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Use an authoritative routable network, travel modes, barriers, time, accessibility, and CRS validation.

Solution 2

Propagate survey/sensor/classification uncertainty and show interval-sensitive ranks.

Solution 3

Define representation, consent, accessible communication, correction, grievance, feedback, and no-retaliation safeguards.

Solution 4

Informality is a legal/social planning context; converting it into a risk score can stigmatize residents and enable punitive misuse.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated